

1. Executive Summary

This report uses only the two retest CSV files supplied by the user. It compares final answer correctness, likely wrong MATCH labels, DAG/operator sequence quality, SPARQL generation quality, scan reliability, and per-query winners.

Main conclusion: Qwen is stronger overall. It has many more MATCH answers, many more MATCH+PARTIAL answers, better head-to-head answer results, and better SPARQL/scan reliability. Llama has a few individual wins but is mostly weaker in this retest.

2. Answer Correctness Counts

Qwen: total 142; MATCH 52 (36.6%); PARTIAL 23 (16.2%); MISMATCH 60 (42.3%); CRASH 7 (4.9%); MATCH+PARTIAL 75 (52.8%).

Llama: total 143; MATCH 12 (8.4%); PARTIAL 13 (9.1%); MISMATCH 112 (78.3%); CRASH 6 (4.2%); MATCH+PARTIAL 25 (17.5%).

Head-to-head on shared question occurrences:

- Tie: 76 (53.5%)
- Qwen: 62 (43.7%)
- Llama: 4 (2.8%)

Status pair counts:

- Qwen MISMATCH vs Llama MISMATCH: 58
- Qwen MATCH vs Llama MISMATCH: 30
- Qwen PARTIAL vs Llama MISMATCH: 21
- Qwen MATCH vs Llama MATCH: 12
- Qwen MATCH vs Llama PARTIAL: 10
- Qwen CRASH vs Llama CRASH: 4
- Qwen CRASH vs Llama MISMATCH: 3
- Qwen PARTIAL vs Llama PARTIAL: 2
- Qwen MISMATCH vs Llama CRASH: 1
- Qwen MISMATCH vs Llama PARTIAL: 1

3. Checking Whether MATCH Is Actually Correct

I created a per-query audit CSV. For JSON/numeric ground truths, expected values must appear in the answer or scan raw rows. For no-data ground truths, answers must clearly say no data/no rows/no rule. Zero answers for no-data ground truth are marked NEEDS_REVIEW because zero is only safe for count questions.

Qwen audit verdicts: WRONG=64, CORRECT=26, UNDER_LABELLED=14, PARTIAL_OR_WRONG=13, PARTIAL_OK=13, NEEDS_REVIEW=12

Llama audit verdicts: WRONG=115, CORRECT=11, NEEDS_REVIEW=9, UNDER_LABELLED=4, PARTIAL_OR_WRONG=3, PARTIAL_OK=1

Potential wrong/review MATCH labels found: 27 total. These rows are listed in

Detailed_Qwen_vs_Llama_Per_Query_Audit.csv.

Review example (Qwen): Across the entire investor database, how many Gold holdings does investor INV-005 hold? | GT: There is no data in raw_data_sample to answer this query. | Answer: Investor INV-005 holds 0 Gold holdings. | Audit: NEEDS_REVIEW.

Review example (Qwen): Across the entire investor database, how many investors affected by the Interest Rate Hike scenario have a ... | GT: There is no business rule/data to calculate this from raw_data_sample. | Answer: No investors affected by the Interest Rate Hike scenario have a Moderate portfolio health status. | Audit: WRONG.

Review example (Qwen): Can you tell me what the average holding duration in days for investor INV-012? | GT: There is no data in raw_data_sample to answer this query. | Answer: A database error prevented data retrieval. | Audit: WRONG.

Review example (Qwen): Can you tell me what the total current value of ETF holdings for investor INV-010? | GT: There is no data in raw_data_sample to answer this query. | Answer: The total current value of ETF holdings for investor INV-010 is 0. | Audit: NEEDS_REVIEW.

Review example (Qwen): Could you show me total Lump Sum investments made per investor across all time.? | GT: (no rows returned) | Answer: Deepak Sharma has made total Lump Sum investments of 35656852 across all time. | Audit: WRONG.

Review example (Qwen): How many Redemption transactions has investor INV-008 made? | GT: There is no data in raw_data_sample to answer this query. | Answer: Investor INV-008 has made 0 Redemption transactions. | Audit: NEEDS_REVIEW.

Review example (Qwen): How many holdings does investor INV-015 have in total? | GT: There is no data in raw_data_sample to answer this query. | Answer: Investor INV-015 has a total of 0 holdings. | Audit: NEEDS_REVIEW.

Review example (Qwen): How many investors affected by the Interest Rate Hike scenario have an At Risk portfolio health status? | GT: There is no business rule/data to calculate this from raw_data_sample. | Answer: No investors affected by the Interest Rate Hike scenario have an At Risk portfolio health status. | Audit: WRONG.

4. DAG / Operator Sequence Strength

DAG quality was scored simply: one point each for Retrieve, Generate, Scan, Validate, and Explain appearing in the planned sequence. This checks pipeline completeness, not semantic correctness.

Qwen: average DAG completeness score 4.81/5.

Llama: average DAG completeness score 4.86/5.

DAG head-to-head winner Tie: 139 (97.9%)

DAG head-to-head winner Llama: 3 (2.1%)

Interpretation: both models usually create plausible DAGs. Qwen does not win by DAG shape alone; it wins because its generated query and answer survive later stages more often.

5. SPARQL / Scan Generation Strength

SPARQL quality was scored from generated query presence, SELECT/PREFIX shape, scan success, no unknown terms, and scan error absence. This captures practical query generation quality better than DAG shape alone.

Qwen: average SPARQL/scan score 4.32/5; scan success 117; scan error 18; rows with unknown terms 15.

Llama: average SPARQL/scan score 2.67/5; scan success 60; scan error 77; rows with unknown terms 78.

SPARQL head-to-head winner Qwen: 79 (55.6%)

SPARQL head-to-head winner Tie: 53 (37.3%)

SPARQL head-to-head winner Llama: 10 (7.0%)

Interpretation: Qwen is much stronger at producing SPARQL that reaches scan success. Llama has many more scan errors and many more rows with unknown RDF terms.

6. Failure Pattern Comparison

Qwen failure stages: blank=67, Validate=51, Scan=17, Exception=7

Llama failure stages: Scan=55, Validate=49, blank=33, Exception=6

Qwen failures are often validation-stage or specific crashes. Llama failures are more heavily scan-stage, meaning the generated SPARQL often cannot execute correctly against the graph.

7. Variant / Repeated Query Check

Qwen: 12 exact repeated questions, 12 extra repeated rows, 0 repeated questions with mixed statuses.

Llama: 12 exact repeated questions, 12 extra repeated rows, 0 repeated questions with mixed statuses.

Repeated queries are internally consistent within each model, but Qwen and Llama often differ on the same repeated query.

8. Representative Query Examples

Qwen answer-winner examples

Question: Across the entire investor database, how many Gold holdings does investor INV-005 hold?

Ground Truth: There is no data in raw_data_sample to answer this query.

Qwen (MATCH, audit NEEDS_REVIEW): Investor INV-005 holds 0 Gold holdings.

Llama (PARTIAL, audit NEEDS_REVIEW): Investor INV-005 holds 0 Gold holdings across the entire investor database.

Question: Across the entire investor database, how many investors affected by the

Interest Rate Hike scenario have a Moderate p...

Ground Truth: There is no business rule/data to calculate this from raw_data_sample.

Qwen (MATCH, audit WRONG): No investors affected by the Interest Rate Hike scenario have a Moderate portfolio health status.

Llama (PARTIAL, audit NEEDS_REVIEW): There are 0 investors affected by the Interest Rate Hike scenario with a Moderate portfolio health status.

Question: Across the portfolio database, list all redemption events for investor inv-008 with dates and amounts.?

Ground Truth: There is no data in raw_data_sample to answer this query.

Qwen (MATCH, audit CORRECT): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MISMATCH, audit WRONG): A database error prevented data retrieval.

Question: Among all investors, which one holds the highest total Gold investment value?

Ground Truth: (no rows returned)

Qwen (MATCH, audit CORRECT): No data was found in the knowledge graph. The requested entity may not exist.

Llama (PARTIAL, audit WRONG): The investor with the highest total Gold investment value is INV001 with a value of 35,656,852.

Llama answer-winner examples

Question: Compare the preferred sector in the investor's profile to the sector with the highest allocation percentage in their ...

Ground Truth: {"investor_id": "INV-001", "stated_sector_focus": "Government, Gold, Debt", "highest_actual_sector": "Priva...

Qwen (CRASH, audit WRONG): ERROR: Operator Compare not found in registry!

Llama (MISMATCH, audit WRONG): No data was found in the knowledge graph. The requested entity may not exist.

Question: Summarise the total cash flow amount for each month for investor INV003.?

Ground Truth: [{"transaction_month": "2023-04", "total_cash_flow": 5000}]

Qwen (MISMATCH, audit WRONG): A database error prevented data retrieval.

Llama (PARTIAL, audit UNDER_LABELLED): For investor INV003, the total cash flow amount for April 2023 is \$5000.

Question: What was the total investment cost made in 2022?

Ground Truth: 30560651

Qwen (CRASH, audit WRONG): ERROR: Request timed out.

Llama (MISMATCH, audit WRONG): The total investment cost made in 2022 was \$0.

Question: Which investors have zero dividend income from all their holdings?

Ground Truth: [{"investor_id": "INV-002", "investor_name": "Neha Sharma 2"}, {"investor_id": "INV-003", "investor_name": ...

Qwen (CRASH, audit WRONG): ERROR: Request timed out.

Llama (MISMATCH, audit WRONG): A database error prevented data retrieval.

Tie answer-winner examples

Question: Across the portfolio database, what was the total cash inflow from deposits and sip contributions in the last 12 months?

Ground Truth: 0

Qwen (MISMATCH, audit WRONG): A database error prevented data retrieval.

Llama (MISMATCH, audit UNDER_LABELLED): No data was found in the knowledge graph. The requested entity may not exist.

Question: Among all investors, which ones have not purchased any new holdings in the last 12 months?

Ground Truth: [{"investor_id": "INV-001", "investor_name": "Deepak Sharma"}, {"investor_id": "INV-002", "investor_name": ...

Qwen (MISMATCH, audit WRONG): A database error prevented data retrieval.

Llama (MISMATCH, audit WRONG): A database error prevented data retrieval.

Question: Among all investors, which ones hold Private Equity investments?

Ground Truth: [{"investor_id": "INV-001", "investor_name": "Deepak Sharma"}]

Qwen (MISMATCH, audit WRONG): No data was found in the knowledge graph. The requested entity may not exist.

Llama (MISMATCH, audit WRONG): A database error prevented data retrieval.

Question: Among all investors, who has the longest average holding duration across all investors?

Ground Truth: {"investor_id": "INV-001", "investor_name": "Deepak Sharma", "average_holding_duration_days": 1359.1}

Qwen (MISMATCH, audit WRONG): A database error prevented data retrieval.

Llama (MISMATCH, audit WRONG): A database error prevented data retrieval.

9. Final Conclusion

Qwen is better for this financial query pipeline. Its strengths are higher answer correctness, more usable partial answers, stronger SPARQL/scan performance, and fewer scan-stage failures. Llama can still produce a plausible DAG, but its SPARQL execution and final factual correctness are much weaker in this retest.

Use the PDF for easy explanation and the per-query audit CSV to check every query one by one.